Lesson 1: Paper Power

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

No preparation needed. Everything is collected with student helpers during the lesson. **No screens** are used. All recorded work is glued into **STEAM journals**, which are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:
To discover that changing the shape of paper can change how strong it is.

Success Criteria:
I can:

- fold and roll paper into different shapes.
- test how many pebbles each shape can hold.
- say which shape was strongest and why.

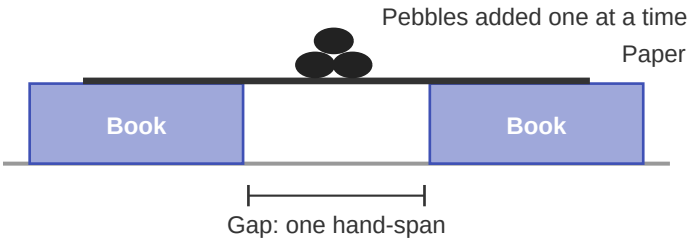
Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
A4 scrap or copy paper	4 sheets per pair, plus 1 per student	3 sheets for testing, 1 for recording
Books (class library)	2 per pair, similar thickness	The two sides of the paper bridge
Black pebbles	About 15 per pair, roughly the same size	The weights
Sticky tape	A short piece per pair	Holding the tube closed
Glue sticks	1 per pair	Gluing the record into journals
STEAM journals	1 per student	Shelves by the door

How the test works

Side view of the test



The 3 shapes (looking from the end)

- Place two books flat on the table, **one hand-span apart**. Keep this gap the same for every test so it is fair.
- Lay the paper across the gap like a bridge.
- Gently place pebbles **one at a time** in the middle. Count each one.
- Stop when the paper sags down to the table or collapses. That number is the score.
- Test 3 shapes: **Flat** (just lay it down), **Fan** (fold back and forth like a paper fan) and **Tube** (roll it up and tape it closed).
- For the fan, the folds must run **from book to book**, not across the gap.
- Pebbles can roll off a tube. If they do, place them in a small pile on top or count the pebble that falls as the last one.

Key words

strong, weak, shape, fold, roll, test, predict, weight

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Read the Learning Intention and Success Criteria aloud.
3 to 10 min	CIC hook	Hold up one sheet of paper. Ask: *"Could this piece of paper hold up these pebbles?"* Thumbs up or down. Lay it flat across a gap between two books and place one pebble in the middle. It sags. Say: *"Paper is weak... or is it? Watch."* Fold a second sheet into a fan in front of them and lay it across the same gap. Add pebbles one by one with the class counting aloud. It holds! Ask: *"It is the same paper. What changed?"* Do not show the tube yet.
10 to 14 min	Explain the challenge	Tell them: *"Today you are paper engineers. You will test 3 shapes: flat, fan and tube."* Show how to roll a tube and tape it. Show how to place pebbles gently one at a time and when to stop. Each pair points to the shape they predict will win. Draw the recording layout (below) on the board.
14 to 30 min	Test in pairs	Choose helpers to hand out: 2 books, about 15 pebbles, 3 sheets and a piece of tape per pair. One partner places pebbles while the other counts, then swap for the next shape. Roam with the questions below.
30 to 38 min	Record	Give each student 1 sheet. They fold it in half and draw the 3 shapes on the front, write how many pebbles each held, and circle the strongest. Helpers hand out journals. Students glue the folded sheet into their journal and write the date and the title **Paper Power**.
38 to 45 min	Share and pack up	3 or 4 pairs share using the sentence stem on the board. Finish with the key idea (below). Pack up: test paper into recycling, pebbles counted back into their container, books back, journals onto the shelves by the door.

Draw this on the board

Flat	Fan	Tube
(draw it)	(draw it)	(draw it)
___ pebbles	___ pebbles	___ pebbles

Circle the strongest.

Sentence stem: **The _____ was strongest because _____.**

Questions to ask while roaming

- What happened when you added more pebbles?
- Is it the same paper? So what changed?
- Why do you think the fan held more than the flat paper?
- Which shape do you think will win next? Why?

Support and extension

Who	What to do
Support	Fold the first fan together with the pair. Let them record with drawings and numbers only. Pair a confident counter with a student who needs help counting.
Extension	Fast finishers (often Year 2) try a different fold, or a fat tube against a thin tube. Can they make a shape that holds even more? Write one sentence explaining why.

Key idea to finish on: Same paper, different shape, different strength. Folding and rolling made the paper stronger **without adding anything new**. Next week we will find out what happens when we twist paper together.

Notes for the teacher

- Any result is fine. Most pairs find the flat sheet holds almost nothing and the fan holds the most. The comparison is the learning, not the exact numbers.
- If a fan collapses straight away, check that the folds run from book to book.
- Pebbles stay on the table and are placed, not dropped. Count them back into their container at pack up.
- Please leave a short note about how far the class got and any pairs who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2C04:** explore the characteristics and properties of materials used to create designed solutions.
- **Maths:** counting a collection, comparing quantities.